

MTH2021 Mathematical Methods 2

Self-study sheet 4

- (a) Prove Morera's theorem.
(b) Extend the Fundamental Theorem of Calculus to integrals in the complex plane.
- Prove Taylor's theorem.
- Use Cauchy's theorem and/or integral formula to evaluate the integral

$$\oint_C \frac{\cos z}{(z+i)(z-2)} dz$$

where the simple closed contour C is:

- the circle of radius $\frac{1}{2}$ centered at $z = 0$
 - the circle of radius $\frac{3}{2}$ centered at $z = 0$
 - a square of side 1 centered at $z = 2$
- Use Cauchy's theorem and/or integral formula to evaluate the integral

$$\oint_C \frac{\cos z}{(z-1)^2} dz$$

where the simple closed contour C is

- the circle of radius 1 centered at $z = 1$
 - a square of side 1 centered at $z = 5 + 5i$
- Use Cauchy's theorem and/or integral formula to evaluate the integrals below when C is the contour $|z| = 2$:

$$(a) \oint_C \sin z \, dz \quad (b) \oint_C \frac{\exp(2z)}{(z-1)^{101}} dz \quad (c) \oint_C \frac{\sin z}{(z-4)^{101}} dz$$

- By applying the Cauchy integral formula for the n -th derivative to the function $\exp(-irz)$ (where r is a real constant) and the contour $|z| = 1$ (the unit circle centered at the origin) show that

$$\begin{aligned} \int_0^{2\pi} \exp(r \sin \theta) \cos(n\theta + r \cos \theta) d\theta &= \frac{2\pi r^n}{n!} \cos(n\pi/2) \\ \int_0^{2\pi} \exp(r \sin \theta) \sin(n\theta + r \cos \theta) d\theta &= \frac{2\pi r^n}{n!} \sin(n\pi/2) \end{aligned}$$

7. Find the first few terms of the Taylor series of the function

$$f(z) = \frac{\sin(z-1)}{z}$$

around the point $z = 1$.

8. Find the Laurent series of the following functions in the specified region

(a)

$$f(z) = \frac{1}{(z+1)^2(z-1)^2} \quad \text{for } 0 < |z-1| < 2$$

(b)

$$f(z) = \frac{1}{(z+1)^2(z-1)^2} \quad \text{for } |z| > 1$$
